EDS 223: Homework 3

Prioritizing Potential Aquaculture

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```
library(sf)
library(terra)
library(tidyverse)
library(tmap)
```

Background

Marine aquaculture consists of breeding, rearing, and harvesting marine animals and plants[1]. Responsibly managed commercial aquaculture has been identified as an environmentally sustainable method for producing protein. As global seafood demand continues to increase, marine aquaculture expansion is being assessed as a economically and resource-efficient way to meet those needs [2]. Constraints for potential marine aquaculture potential include depth, dissolved oxygen, phytoplankton availability, shipping traffic, oil rigs, and marine protected areas [3]. The U.S. Exclusive Economic Zone (EEZ) extends no more than 200 nautical miles from the U.S. territorial sea baseline, and the U.S. has sovereign rights to explore, exploit, conserve, and manage natural resources within that boundary [4]. This analysis examines the potential for developing marine aquaculture for several species of oysters and Pacific calico scallop in the West Coast Exclusive Economic Zones.

Data

Temperature and depth suitable ranges for the Pacific calico scallop were obtained from the SeaLifeBase website.

Sea surface temperature (SST) data, annual averages in Kelvin, for the years 2008-2012 was originally generated from NOAA's 5km Daily Global Satellite Sea Surface Temperature Anomaly v3.1.

Bathymetry data was obtained from the General Bathymetric Chart of the Oceans (GEBCO).

Boundary data for the West Coast Exclusive Economic Zone was obtained from Marineregions.org.

Methods

Sea surface temperature data rasters were stacked, reprojected to match bathymetry crs, averaged, and converted from Kelvin to degrees Celcius. The bathymetry raster was cropped and resampled to match the extent, resolution, and alignment of the mean SST raster. EEZ region polygons were rasterized using region ID to match the mean SST raster. Raster cell area was calculated, converted to square kilometers, and cell area values were masked to EEZ boundaries.

A function was created to calculate and map suitable EEZ habitat for oysters and the Pacific calico scallop, using min/max SST, min/max depth and species name as arguments. The reclassifications were then applied to the SST and depth rasters, assigning suitable cells a value of 1 and unsuitable cells 0. The two suitability rasters were then multiplied together, and only cells that met both SST and depth suitability remained assigned as 1. Unsuitable cells were converted to NA and suitable cells were multipled by the cell area. Suitable habitat area was then summed within each EEZ region and joined with the EEZ shapefile. Total suitable area was mapped by EEZ region.

```
# Reading in rasters and shapefile

annual_temp_08 <- rast("data/average_annual_sst_2008.tif")

annual_temp_09 <- rast("data/average_annual_sst_2009.tif")

annual_temp_10 <- rast("data/average_annual_sst_2010.tif")

annual_temp_11 <- rast("data/average_annual_sst_2011.tif")

annual_temp_12 <- rast("data/average_annual_sst_2012.tif")

depth <- rast("data/depth.tif")

regions <- st_read("data/wc_regions_clean.shp")
```

```
# Stacking temperature rasters
temp_stack <- c(annual_temp_08, annual_temp_09, annual_temp_10, annual_temp_11,
annual_temp_12)
```

```
# Checking if coordinate reference systems match
if (st_crs(regions) == st_crs(depth)) {
  print("Coordinate reference systems match")
} else {
  warning("Updating coordinate reference system to match")

   # Transform EEZ regions (vector) to match depth CRS,
```

```
  regions <- st_transform(
    regions,
    crs(depth)
  )
}
```

```
[1] "Coordinate reference systems match"
```

```
# Check if coordinate reference systems match
if (crs(temp_stack) == crs(depth)) {
  print("Coordinate reference systems match")
} else {
  warning("Updating coordinate reference system to match")

  # Transform raster to match depth CRS
  temp_stack <- project(
    temp_stack,
    crs(depth)
  )
}
```

```
Warning: Updating coordinate reference system to match
```

```
# Mean SST from 2008-2012
mean_sst <- app(temp_stack, mean, na.rm = TRUE)

# Convert mean SST from Kelvin to Celsius
mean_sst <- mean_sst - 273.15

# Crop depth to SST extent
depth_crop <- crop(depth, mean_sst)

# Resample cropped depth to match SST resolution and alignment
depth_resampled <- resample(
  depth_crop,
  mean_sst,
  method = "bilinear"
)
```

```r
# Checking resolution, extent, and crs
res(depth_resampled)
```

```
[1] 0.04165905 0.04165905
```

```r
res(mean_sst)
```

```
[1] 0.04165905 0.04165905
```

```r
ext(depth_resampled)
```

```
SpatExtent  :  -131.98475232999999,  -114.9878608010912,  29.992079988813192,
49.988422964000002 (xmin, xmax, ymin, ymax)
```

```r
ext(mean_sst)
```

```
SpatExtent  :  -131.98475232999999,  -114.9878608010912,  29.992079988813192,
49.988422964000002 (xmin, xmax, ymin, ymax)
```

```r
crs(depth) == crs(mean_sst)
```

```
[1] TRUE
```

```r
# Check if depth and SST can be stacked
depth_sst <- c(depth_resampled, mean_sst)
```

```r
# Rasterizing EEZ regions using region IDs
region_rast <- terra::rasterize(regions, mean_sst, field = "rgn_id")

# Calculating cell area as km^2
cell_area <- cellSize(mean_sst) / 1e6

# Masking cell sizes to EEZ regions
eez_cell_area<- mask(cell_area, regions)
```

```r
# Function for creating mapping suitable Exclusive Economic Zones for Aquaculture
suitable_eez <- function(min_sst, max_sst, min_depth, max_depth, species_name){

# Reclassifying SST for species suitability
sst_rcl <- matrix(
  c(
    -Inf, min_sst, 0,
    min_sst, max_sst, 1,
    max_sst, Inf, 0
  ),
  ncol = 3,
  byrow = TRUE
)

# Reclassify sst based on species sst suitability range
sst_stbl <- classify(mean_sst, rcl = sst_rcl)


# Reclassifying depth for species suitability
depth_rcl <- matrix(
  c(
    -Inf, min_depth, 0,
    min_depth, max_depth, 1,
    max_depth, Inf, 0
  ),
  ncol = 3,
  byrow = TRUE
)

# Reclassify depth based on species depth suitability range
depth_stbl<- classify(depth_resampled, rcl = depth_rcl)

# Combining suitability rasters
habitat_stbl <- lapp(c(sst_stbl, depth_stbl), fun=function(sst, depth)sst *
depth)

names(habitat_stbl) <- "suitability"

# Changing cells less than 1 to NA
habitat_stbl[habitat_stbl < 1] <- NA

# Calculating area for each suitable habitat cell
area_stbl <- habitat_stbl * eez_cell_area
```

```r
# Summing suitable habitat area for each EEZ region
total_suitable_area <- zonal(
  area_stbl,
  region_rast,
  fun = "sum",
  na.rm = TRUE)|>
  rename(suitable_area_km2 = suitability)

# Joining total suitable habitat to EEZ regions
regions_area <- regions |>
  left_join(
    total_suitable_area,
    by = "rgn_id"
  )

# Print suitable area values for reference
print(
  regions_area |>
    st_drop_geometry() |>
    select(rgn, suitable_area_km2)
)

# Creating map of suitable area by EEZ region
habitat_map <-
  tm_basemap("Esri.NatGeoWorldMap", zoom = 5, alpha = 0.55) +
  tm_shape(regions_area) +
  tm_polygons(
    fill = "suitable_area_km2",
    fill.scale = tm_scale_continuous(
      values = c("#f6eff7", "#bdc9e1", "#67a9cf", "#1c9099", "#016c59")
    ),
    fill.legend = tm_legend(
      title = "Suitable Area (km²)",
      position = tm_pos_out("right", "center")
    ),
    col = "black",
    lwd = 1
  ) +
  tm_compass(position = c("left", "top"), size = 1.5) +
  tm_scalebar(position = c("left", "bottom"), width = 9) +
  tm_title(
    paste(species_name, "Suitable Area by EEZ Region"),
```

```
    position = tm_pos_out(
      "center",
      "top",
      pos.h = "center"
    ), size = 1
  ) +
  tm_layout(
    inner.margins = c(0.08, 0.10, 0.02, 0.02),
    component.autoscale = FALSE
  )

return(habitat_map)
}
```

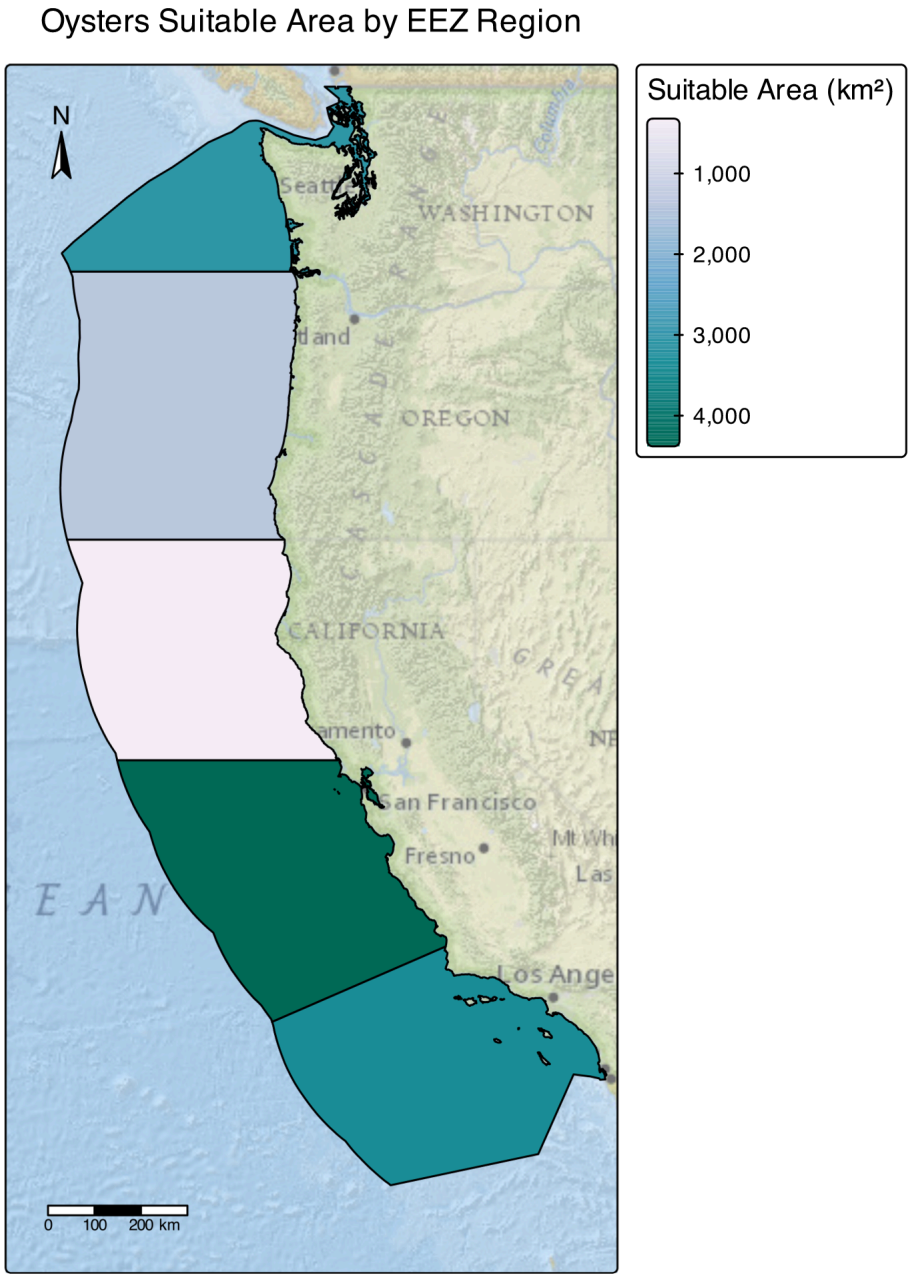
Results

Oysters

The Central California EEZ region had the highest amount of total suitable habitat at approximately 4,390 km², followed by Southern California at 3,348 km² and Washington at 3,063 km². The regions with the lowest total suitable habitat were Oregon and Northern California, which contained 1,381 km² and 340 km².

```
suitable_eez(min_sst = 11, max_sst = 30, min_depth = -70, max_depth = 0,
species_name = "Oysters")
```

```
                rgn suitable_area_km2
1            Oregon         1380.6397
2 Northern California          340.3413
3  Central California         4389.6164
4 Southern California         3347.7995
5        Washington         3062.5924
```



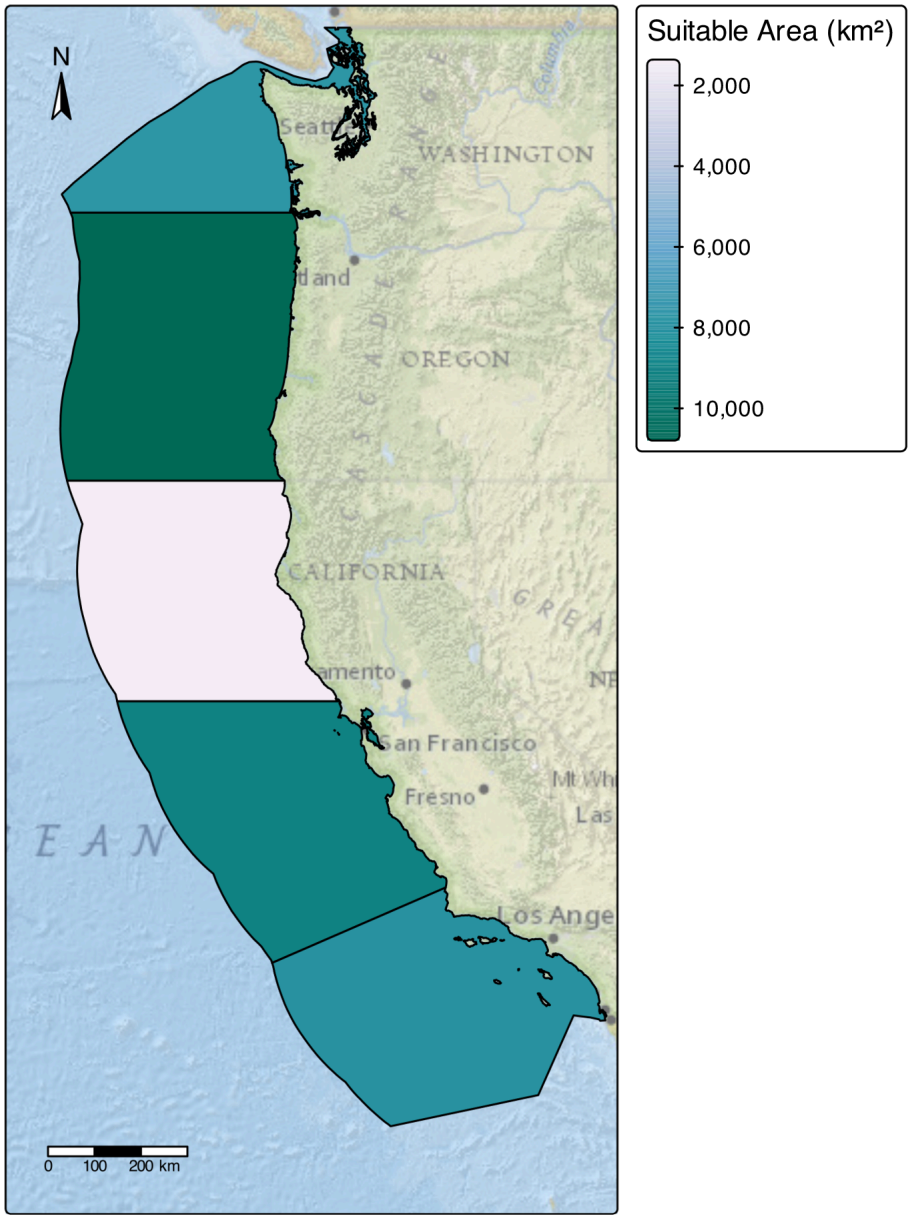
Pacific calico scallop

Oregon had the highest suitable area at approximately 10,799 km², followed by Central California at 9,113 km². Southern California and Washington had similar amounts of suitable habitat, with 7,986 km² and 7,815 km², respectively. Northern California had the lowest total suitable habitat at 1,367 km².

```
suitable_eez(min_sst = 11, max_sst = 28.5, min_depth = -180, max_depth = 0,species_name = "Pacific calico scallop")
```

```
                 rgn suitable_area_km2
1             Oregon         10799.145
2 Northern California          1366.504
3  Central California          9113.274
4 Southern California          7986.342
5         Washington          7815.061
```

Pacific calico scallop Suitable Area by EEZ Region



Conclusion

Suitable aquaculture habitat varied among EEZ regions and between oysters and Pacific calico scallops. Central California had the highest total suitable area for oysters, and Oregon had the highest total suitable area for Pacific calico scallops. Pacific calico scallops had more total suitable habitat across all regions compared to oysters. Northern California had the lowest total suitable area for both oysters and Pacific calico scallops. The scallop depth range was 0–180 m compared

to 0–70 m for oysters, which likely contributed to the greater amount of suitable habitat. The temperature ranges were more similar, 11–30 °C for oysters and 11–28.5 °C for Pacific calico scallops, and any limitations from temperature may similarly have affected the suitability for both species. This analysis shows how species-specific habitat requirements can limit or expand the areas within EEZ regions that may have potential for aquaculture.

References

[1] NOAA Fisheries. (n.d.). *Aquaculture*. National Oceanic and Atmospheric Administration. https://www.fisheries.noaa.gov/topic/aquaculture [Accessed November 7, 2026]

[2] National Centers for Coastal Ocean Science. (2021, October 19). *Aquaculture opportunity areas for federal waters of the United States Exclusive Economic Zone*. National Oceanic and Atmospheric Administration. https://coastalscience.noaa.gov/project/aquaculture-opportunity-areas-for-federal-waters-of-the-united-states-exclusive-economic-zone/ [Accessed November 7, 2026]

[3] Gentry, R. R., Froehlich, H. E., Grimm, D., Kareiva, P., Parke, M., Rust, M., Gaines, S. D., & Halpern, B. S. (2017). *Mapping the global potential for marine aquaculture. Nature Ecology & Evolution, 1*, 1317–1324. https://doi.org/10.1038/s41559-017-0257-9

[4] National Oceanic and Atmospheric Administration. (2026, September 23). *What is the EEZ?* National Ocean Service. https://oceanservice.noaa.gov/facts/eez.html [Accessed November 7, 2026]